

Durable agents for application maintenance

Target use cases and proposed design

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CVE remediation across applications

An advisory affects dependencies in **several applications**.

1. Identify affected applications and start remediation for each.
2. Coordinate each application's fix and track the result.
3. Continue handling other requests while individual fixes run.

Success: every affected application has a traceable outcome.
A reviewed lesson from one fix can inform another.

Pull-request review over time

A review bot returns when a pull request opens or changes.

- Apply approved repository conventions and prior lessons.
- Keep context for reviewing follow-up revisions.
- Work independently from migration or maintenance activity.

Success: review resumes when needed, with shared knowledge and its own working context.

Work triggered by a Jira issue

When an issue is created, start the appropriate application work.

- Retain the request until handled, even if delivery is repeated.
- Work directly or start a finite task and follow its result.
- Let later events or human messages continue the work.

Success: an accepted request survives a lost connection or restart.

Requirements shared by these use cases

- **Continuity:** identity and context outlive a pod.
- **Durable delivery:** accepted inputs remain actionable.
- **Delegation:** a session can keep working while children run.
- **Shared knowledge:** reviewed lessons inform independent workloads.
- **History:** requests, outcomes, and transcripts remain inspectable.

The first version sleeps **between completed inputs**.

The proposed resources

Resource	Purpose
AgentSession	Continuing identity, conversation, and workspace
AgentInput	One durable request addressed to one session
AgentLearning	A reviewable lesson with immutable content and applicability

AgentRun and **AgentWorkflowRun** remain finite invocations.

Controller, session worker, agent runtime

Responsibility	Owner
Watch resources and wake compute	**Controller**
Claim inputs, prepare learnings, persist progress	**Session worker in the harness**
Reason, use tools, selectively read prepared memory	**Agent runtime**

Fetching the next input and refreshing learnings happen **outside the agent invocation**.

Processing one input

1. **Claim** the input and inspect saved progress.
2. **Pin** learnings for new work; restore the snapshot for a retry.
3. **Prepare** memory files, context, and workspace.
4. **Invoke or resume** the agent runtime.
5. **Persist** required results and session state.
6. **Complete** the input, then take the next one or sleep.

Taking an item off the worker queue does **not** prove completion.

Delegation and subsequent inputs

Session activity	Delegated run
Input A: record and dispatch a CVE fix	Created with a stable identity
Complete the dispatch input	Continues executing
Input B: handle another request	Still executing
Input C: process the child outcome	Completed

Recovery must find the **existing child**, not create another one.

Learning selection boundaries

Consumer	When the set is pinned
AgentRun	At creation, before any queueing delay
AgentWorkflowRun	At creation; every stage inherits the same set
AgentSession	Before each new input, including while already awake

Retries retain the original snapshot.

Immutable learning lifecycle

Propose → review → accept or reject

- Approval covers the exact content **and applicability**.
- An accepted replacement changes **future selections**.
- Existing consumers keep their pinned revisions.

> Revocation is refused until every unfinished referencing consumer has finished or actually stopped/cancelled.

Memory files and selective delivery

```
memory/index.md                    summaries and paths
memory/learnings/<revision>.md     full learning content
memory/manifest.json               identities and digests
```

Load a concise approved index and essential guidance automatically.
Read longer procedures and supporting evidence when relevant.

Read-only projection of the pinned set. No MCP required.

Learning delivery evidence

Record	What it establishes
Available	The selected revision was materialized
Advertised	Its summary entered the model's context
Delivered	Full content or an identified excerpt reached the model
Agent-reported consideration	The agent explicitly reported applicability or conflict

A file read does not prove the agent followed the learning.

What survives compute replacement

- Accepted inputs, saved decisions, and action results
- The input's pinned learning snapshot
- Session context and recoverable workspace state
- Correlated history and transcripts while their AgentRuns exist

> Unknown external outcome: reconcile or surface an interruption. Do not blindly repeat the action.

Proposed spike

One complete path: accept → claim → prepare → invoke → persist → sleep/wake

- Prove per-input refresh and stable retry snapshots.
- Exercise delegation while the session handles later work.
- Interrupt claims, effects, completion, and idle shutdown.
- Test queued/workflow consumers against replacement and revocation.
- Qualify real-runtime memory delivery and retained transcripts.

Return reproducible evidence, API/storage proposals, and a go/no-go recommendation.

Review decisions

Review now

Target use cases, resource boundaries, worker ownership, snapshot timing, and the immutable learning lifecycle.

Resolve through the spike

Storage and transaction mechanics, API conditions, ordering/interruption, memory budgets, runtime recovery, and observation coverage.

IDE/CLI continuation and future execution backends remain follow-ups.
The first version does not require VM suspension.

Design and research references

- **ADR 0024:** Durable Agent Sessions, Inputs, and Learnings
- **Spike plan:** `docs/spikes/durable-agent-sessions-and-learnings.md`
- DeepAgents: memory files and selective loading
- Anthropic: The components of agent memory
- Cloudflare: conversation state and memory

Full evidence and limitations are in `docs/research/`.